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ORIGINAL WORK

Seizures and Cognitive Outcome After Traumatic Brain Injury: A Post Hoc Analysis

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Abstract

Introduction: Seizures and abnormal periodic or rhythmic patterns are observed on continuous electroencephalography monitoring (cEEG) in up to half of patients hospitalized with moderate to severe traumatic brain injury (TBI). We aimed to determine the impact of seizures and abnormal periodic or rhythmic patterns on cognitive outcome 3 months following moderate to severe TBI.

Methods: This was a post hoc analysis of the multicenter randomized controlled phase 2 INTREPID[2566] clinical trial conducted from 2010 to 2016 across 20 United States Level I trauma centers. Patients with nonpenetrating TBI and postresuscitation Glasgow Coma Scale scores 4–12 were included. Bedside cEEG was initiated per protocol on admission to intensive care, and the burden of ictal-interictal continuum (IIC) patterns, including seizures, was quantified. A summary global cognition score at 3 months following injury was used as the primary outcome.

Results: 142 patients (age mean + / − standard deviation 32 + / − 13 years; 131 [92%] men) survived with a mean global cognition score of 81 + / − 15; nearly one third were considered to have poor functional outcome. 89 of 142 (63%) patients underwent cEEG, of whom 13 of 89 (15%) had severe IIC patterns. The quantitative burden of IIC patterns correlated inversely with the global cognition score ($r = -0.57$; $p = 0.04$). In multiple variable analysis, the log-transformed burden of severe IIC patterns was independently associated with the global cognition score after controlling for demographics, premorbid estimated intelligence, injury severity, sedatives, and antiepileptic drugs (odds ratio 0.73, 95% confidence interval 0.60–0.88; $p = 0.002$).

Conclusions: The burden of seizures and abnormal periodic or rhythmic patterns was independently associated with worse cognition at 3 months following TBI. Their impact on longer-term cognitive endpoints and the potential benefits of seizure detection and treatment in this population warrant prospective study.

Keywords: Traumatic brain injury, Continuous electroencephalography, Neurocritical care, Cognition, Outcome

Introduction

Traumatic brain injury (TBI) is endemic and affects nearly 3 million each year in the United States alone; nearly one in ten are hospitalized [1]. After brain trauma, the incidence of seizures ranges between 2.6 and 33% [2, 3], and the majority are only detected during continuous electroencephalography (cEEG) monitoring. In patients undergoing cEEG, abnormal periodic or rhythmic patterns that lie along an ictal-interictal continuum (IIC) may be observed in nearly half [3, 4]. The presence of seizures and periodic discharges after brain injury has been associated with injury severity [3], increases in intracranial pressure [5], metabolic crisis [6], and hippocampal atrophy [7]. Brain Trauma Foundation guidelines suggest preventing early seizures in the setting of the most severe TBI

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[8], and experts recommend monitoring and treating seizures in this population [9].

In a recent post hoc analysis of the prospective randomized controlled INTREPID^{2566} trial, seizures were uncommon despite protocolized cEEG monitoring starting within 8 h of injury. Those with severe IIC patterns, including seizures, showed no difference in the level of functional recovery 3 months after injury compared with those without severe IIC patterns [3]. Given the pathophysiological effects of seizures [6, 10], one plausible explanation is that seizures create neuronal injury that impacts *higher* cognitive functions than those captured by functional outcome assessments, such as the Glasgow Outcome Scale-Extended (GOSE). It is well recognized that this may occur in patients with epilepsy [11], but little data exist on the effects of *acute* seizures on cognitive outcome after brain injury. Therefore, we investigated whether the burden of seizures and periodic or rhythmic discharges impacts cognitive recovery at 3 months following moderate to severe TBI.

Methods

We performed a follow-up post hoc analysis of INTREPID^{2566} as previously described [3]. INTREPID^{2566} was a multicenter randomized controlled phase 2 clinical trial of patients with moderate to severe TBI conducted from April 2010 to January 2016 across 20 United States centers. INTREPID^{2566} was conducted to evaluate the safety and efficacy of glycyl-L-2-methylprolyl-L-glutamic acid (trofinetide; NNZ-2566), a novel analog of the tripeptide glycine-proline-glutamate (GPE), a naturally occurring protein within the brain that results from the enzymatic cleavage of insulin-like growth factor-1. Patients were centrally randomly assigned 2:1 to study drug or placebo; those randomly assigned to receive study drug were assigned one of three weight-based dosing tiers. Allocation was masked to participants, clinical care providers, and outcome assessors. Eligible patients were 18–70 years of age with nonpenetrating TBI, postresuscitation Glasgow Coma Scale (GCS) score 4–12, postresuscitation hemodynamic stability, and able to be randomly assigned within 8 h of injury. Exclusion criteria were spinal cord injury, significant bodily coinjuries, prior brain injury requiring hospitalization, severe comorbidities, weight > 150 kg, fluid resuscitation > 6 l prior to randomization, and those at risk for QT prolongation. INTREPID^{2566} demonstrated no significant differences in either adverse events or its primary end point, the GOSE at 3 months.

Standard Protocol Approvals, Registrations, and Patient Consents

INTREPID^{2566} (ClinicalTrials.gov ref: NCT00805818) was conducted in accordance with Good Clinical Practice and was approved for exemption from informed consent (EFIC) by the institutional review board at each participating institution. Legally authorized representatives provided written informed consent in all cases in which they were available prior to initiation of study procedures; patients could be randomly assigned under EFIC until written informed consent could be obtained.

Data Collection

Clinical features were collected, including demographics, TBI-specific parameters (i.e., injury description, Injury Severity Score [ISS], Marshall computed tomography [CT] classification within the first 24 h), neurological examination findings (i.e., postresuscitation GCS, pupillary reactivity), and the use of sedative agents and antiepileptic drugs (AEDs). Per study protocol, sites were encouraged to follow current international and hospital guidelines; no specific recommendations for AED prophylaxis were provided. Variables included in the International Mission for Prognosis and Analysis of Clinical Trials in TBI (IMPACT) score model were specifically recorded, and a sum score was calculated [12]. Functional outcome was documented using GOSE [13] and dichotomized as unfavorable or poor (GOSE 1–4) vs. favorable or good (GOSE 5–8).

Electroencephalography and Seizure Burden

The trial protocol included bedside cEEG initiated on admission to the intensive care unit when possible and continued at least through the 72-h maintenance drug infusion. Thirteen scalp electrodes were positioned based on the international 10–20 System: Fp1, F7, C3, T3, T5, O1, Cz, Fp2, F8, C4, T4, T6, and O2. electroencephalography (EEG) was prospectively interpreted by a Core EEG laboratory consisting of a group of five board-certified EEG readers blinded to treatment allocation at the University of Cincinnati (MP). Post hoc raw cEEG was independently reviewed and quantified by study authors (HL, BF, MAM) trained in the ACNS Standardized Critical Care EEG Terminology [14], as described previously [3]. Seizures and abnormal periodic or rhythmic patterns were classified based on their potential association with seizure activity or neuronal injury; together these classifications were termed the IIC. Electrographic seizures (ESz) were defined as any spikes, sharp waves, or sharp and slow wave complexes with a frequency > 2.5 Hz lasting for 10 s, or any rhythmic or periodic discharges (i.e., IIC patterns) with clear evolution in frequency, morphology, or location. The primary exposure for this study focused on patterns classified as *severe IIC*, defined as ≥ 1.5 Hz lateralized rhythmic delta activity or generalized periodic discharges and any lateralized periodic discharges or ESz. We calculated the IIC *burden* based on IIC prevalence and frequency (e.g., if 1.5-Hz periodic discharges were detected for 50% of an 8 h record, then 25% of the next 24 h, the IIC duration would be 4 h + 6 h = 10 h, and the IIC burden would be 1.5 Hz × 10 h = 15 Hz-hours) and adjusted for recording duration. The IIC burden was calculated for severe IIC patterns only for the purposes of this study.

Cognitive Outcome Measures

Follow-up occurred in person at 3 months following injury, and a neuropsychological battery was performed by trained study personnel at each site blinded to study drug allocation per study protocol. To estimate premorbid intelligence, The North American Adult Reading Test (NAART) [15] was used, which asks patients to read a list of phonetically irregular words. The ability to correctly pronounce even unfamiliar words correlates significantly with verbal and full-scale intelligence independent of age, education, and socioeconomic factors and is thought to be insensitive to the effects of brain injuries [16, 17], although reading performance tasks may underestimate premorbid intelligence after severe TBI [18]. Our primary outcome measure consisted of the Repeatable Battery for the Assessment of Neuropsychological Status (RBANS) [19, 20], which is a brief written test of five major cognitive domains using abbreviated versions of standard tests, such as list learning and story memory (immediate recall), figure copy and line orientation (visuospatial ability), picture naming and semantic fluency (language), digit span and coding (attention), and list recall and story recall (delayed memory). The sum of each of the subscores forms the total, or global cognition score, which ranges from 40 to 160 and has a population age-adjusted mean +/− standard deviation (SD) of 100 +/− 15.

In addition to our primary outcome, we considered each RBANS domain-specific subscore individually. In order to test executive function specifically, we used the Trail Making Test (TMT) parts A and B, which asks the patient to connect sequences of numbers or letters and measures the time and number of errors made during each part.

Statistical Analysis

Clinical variables were summarized using means (+/− SD) or medians (interquartile range [IQR]) for continuous variables as appropriate and proportions for discrete variables. The burden of severe IIC patterns was positively skewed and therefore was log-transformed to approximate a normal distribution (Supplementary Fig. 1). Patients were dichotomized into below-median vs. above-median cognitive outcome based on the median 3 month global cognition score. *t* tests, Wilcoxon, and χ^2 tests were used to compare dichotomous cognitive outcome groups (below-median vs. above-median).

A general linear model was constructed using backward selection with a prespecified significance level to stay of 0.05, with the treatment variable (study drug vs. placebo) included in each model. To avoid overfitting, the significance level based model selection process terminates when addition of a variable to the model increases the predicted residual sum of squares. The variance inflation factor was used to verify variables exhibited no collinearity using a threshold of 10. In an exploratory analysis, we conducted the same analyses with each of the RBANS domain-specific subscores and the TMT. Coefficients from each model were then standardized by subtracting the mean and dividing by twice the corresponding standard deviations so that the resulting coefficients between continuous and untransformed binary predictors were directly comparable [21]. The standardized regression coefficients were then plotted on the same forest plot for each outcome variable for both the full and reduced model. All analyses were conducted using SAS version 9.4 (SAS Institute, Cary, NC), and the plots were generated using R software [22] version 3.6.1.

Data Availability Statement

Data for this study were made available by Neuren Pharmaceuticals Limited through a data use agreement. Neuren Pharmaceuticals Limited retains sole ownership of individual participant data.

Results

A total of 261 patients with moderate to severe TBI were eligible for enrollment in the INTREPID2566 study; 10 were excluded prior to study drug administration (Fig. 1, Table 1, and Supplementary Table 1). At 3 months following injury, 142 of 251 (57%) patients survived and were able to perform cognitive testing; 63 of 251 (25%) survived to assessment could not complete cognitive testing, 24 (10%) died, and 15 (6%) were lost to follow-up (Supplementary Table 2). Patients who were able to complete cognitive testing were younger (32 +/− 13 years vs. 38 +/− 15 years; $p < 0.01$) and more likely men (131 of 142 [92%] vs. 90 of 109 [83%]; $p = 0.03$) with better admission GCS (median 7 [IQR 6–9] vs. median 7 [IQR 6–8]; $p < 0.01$), lower ISS (median 22 [IQR 13–30] vs. median 26 [IQR 18–34]; $p = 0.03$), and lower IMPACT sum scores (median 2 [IQR 0–4] vs. median 3 [IQR 2–6]; $p < 0.01$). On CT, patients able to perform cognitive testing had less traumatic interventricular hemorrhage (IVH)

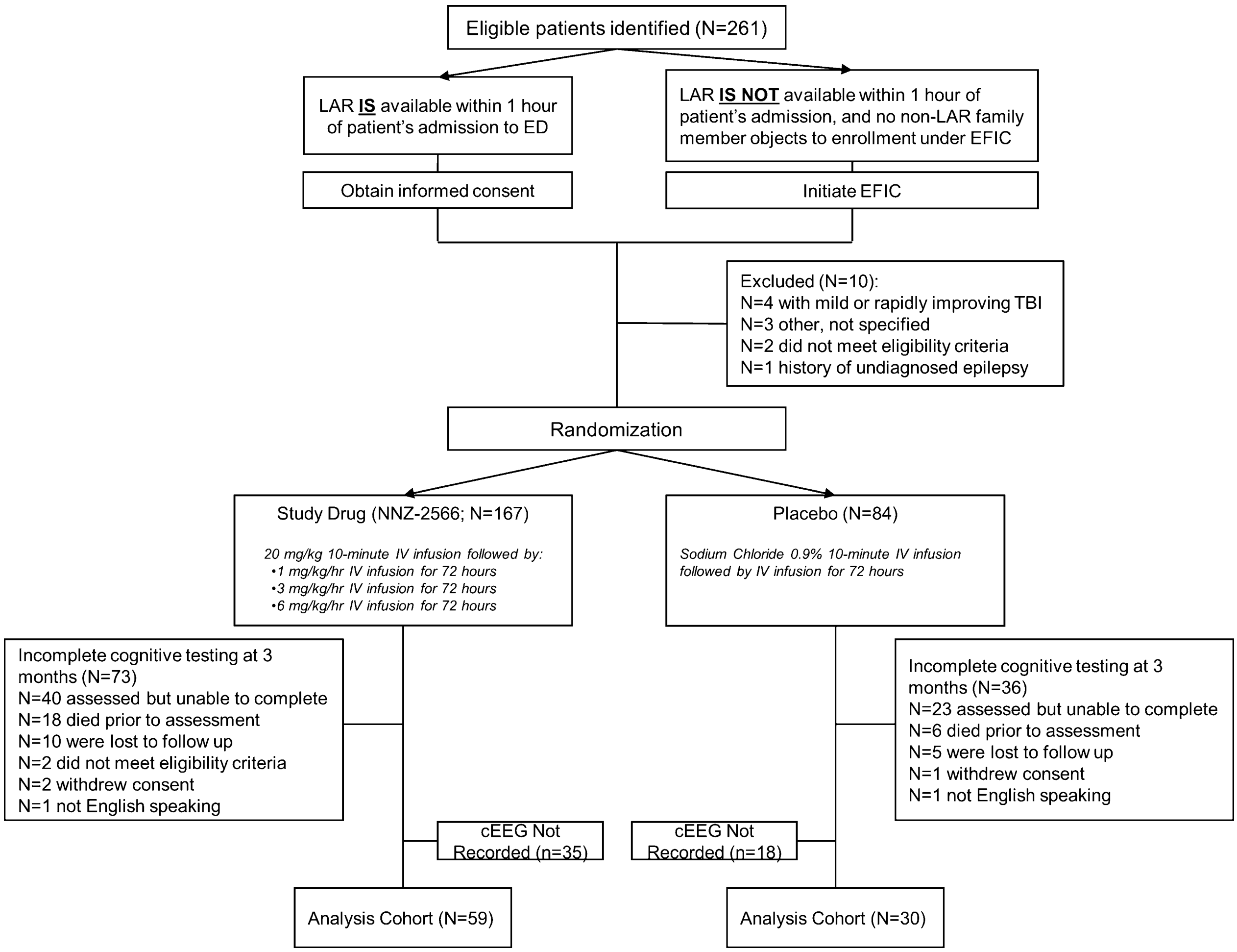

Fig. 1 CONSORT diagram. Modified from study protocols (Neu-2566-TBI-001 and Neu-2566-TBI-002). cEEG, continuous electroencephalography, EFIC, exception from informed consent, ED, emergency department, LAR, legally authorized representative

Table 1 EEG variables and cognitive outcome

Variable	All patients with EEG & cognitive assessment; $n=89$	Below-median cognitive outcome[a]; $n=44$	Above-median cognitive outcome; $n=45$	p
IIC pattern present, n (%)	13 (14.6)	9 (20.5)	4 (8.9)	0.21
IIC Burden, mean +/− SD	0.37 +/− 1.7	0.73 +/− 2.4	0.03 +/− 0.13	0.06
Background within first 24 h (delta predominant), n (%)	30 (33.7)	12 (27.3)	18 (40.0)	0.30
Posterior dominant rhythm within first 24 h (absent), n (%)	67 (75.3)	33 (75.0)	34 (75.6)	1.00
Sleep transients within first 24 h (absent), n (%)	65 (45.8)	31 (47.7)	34 (44.2)	0.76
Worst background anytime during EEG (delta predominant, discontinuous, or suppressed), n (%)	45 (50.6)	20 (45.5)	25 (55.6)	0.46
Discontinuous background anytime during EEG, n (%)	10 (11.2)	6 (13.6)	4 (8.9)	0.71

EEG, electroencephalography, IIC, ictal-interictal continuum, SD, standard deviation

[a] Below-median cognitive outcome dichotomized based on global cognitive score below the median for the entire cohort

(31 of 142 [22%] vs. 40 of 109 [37%]; $p < 0.01$). Of patients able to perform cognitive testing, 43 of 140 (31%) had poor 3-month functional outcome; two patients did not have available functional outcome assessment.

Univariate Associations with Cognitive Outcome

The mean global cognition score across the cohort was 80.5 + / − 15.3 (Supplementary Fig. 2a). There were no significant group differences between mean global cognition scores and GOSE (Supplementary Fig. 2b; $p = 0.42$). The mean time to complete the TMT parts A and B were 46.7 + / − 38.0 s and 113.4 + / − 78.1 s, respectively (Table 2). There were no significant differences between cognitive testing and treatment allocation vs. placebo (Supplementary Table 3). Cognitive outcome was dichotomized as above-median vs. below-median based on a median global cognition score of 80 (IQR 73–92). Statistically significant univariate associations with cognitive outcome included ethnicity and estimated premorbid intelligence.

A total of 89 of 142 (63%) patients underwent EEG monitoring starting a mean of 13.6 + / − 10.4 h following injury, and a median of 64 (IQR 40–93) hours were evaluated. A total of 13 of 89 (15%) had severe IIC patterns, all within 72 h and with focal distribution on scalp EEG. Background features, including the presence or absence of anterior–posterior organization and sleep transients, were not associated with cognitive outcome (Table 1). The presence of severe IIC patterns was not significantly more common in those with below-median cognitive outcome (9 of 44 [20.5%]) compared with those with good cognitive outcome (4 of 45 [8.9%]; $p = 0.21$). Although there was greater IIC burden in those with below-median outcome, this did not reach the threshold for significance (0.73 + / − 2.4 Hz-hr vs. 0.03 + / − 0.13 Hz-hr in those with above-median outcome; $p = 0.06$). However, in the 13 of 89 (15%) who had severe IIC patterns, the log of the IIC burden was inversely correlated with the global cognition score ($r = -0.57$; $p = 0.04$; Fig. 2).

Adjusted Associations with Cognitive Outcome

In a multiple variable regression model, including $n = 89$ with both cognitive testing and EEG data, the log transform of the burden of severe IIC patterns was independently associated with the global cognition score after controlling for age, sex, ISS, postresuscitation GCS and

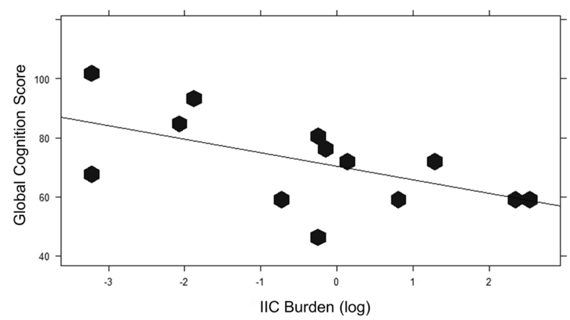

Fig. 2 Relationship between the burden of ictal-interictal continuum patterns and global cognition scores. A scatterplot of the relationship between the global cognition score (*y*-axis) and the log-transformed burden of severe ictal-interictal continuum (IIC) patterns (*x*-axis) across the 13 patients with IIC patterns. There was a significant inverse correlation (Pearson $r = -0.57$; $p = 0.04$)

Table 2 Results of cognitive testing in those with EEG

Variable	Below-median cognitive outcome, $n = 44$		Above-median cognitive outcome, $n = 45$		Pooled difference, mean (95% confidence interval of the mean)
	Value, mean + / − SD	Range	Value, mean + / − SD	Range	
Global Cognition Score	67.1 + / − 10.9	42–78	91.6 + / − 9.1	80–111	24.5 (20.3–28.7)*
Index Scores					
Immediate Recall	69.0 + / − 13.9	40–90	94.7 + / − 13.7	65-–23	25.7 (19.9–31.5)*
Visuospatial Construction	78.5 + / − 15.1	50–112	96.5 + / − 15.7	64–126	18.0 (11.5–24.5)*
Language	76.3 + / − 17.6	40–112	91.7 + / − 11.8	74–124	15.4 (9.1–21.7)*
Attention	74.3 + / − 15.3	43–112	92.8 + / − 14.6	56–125	18.5 (12.2–24.8)*
Delayed Memory	67.1 + / − 19.2	40–97	95.6 + / − 11.6	64–122	28.5 (21.8–35.1)*
TMT A Time (s)	63.2 + / − 55.3	20–300	33.8 + / − 12.5	14–68	29.4 (12.6–46.2)*
TMT A Errors	0.14 + / − 0.55	0–3	0.20–0.50	0–2	0.06 (− 0.16 to 0.29)
TMT B Time (s)	146.6 + / − 101.7	48–542	85.2 + / − 39.7	36–253	61.4 (29.0–93.9)*
TMT B Errors	1.4 + / − 2.2	0–9	0.7 + / − 1.1	0–5	0.69 (0.06–1.43)

EEG, electroencephalography, TMT, Trail Making Test

*$p < 0.001$

pupillary reactivity, radiologic findings, the use of AEDs, the use of γ-aminobutyric acid (GABA)-acting sedative agents, and background EEG characteristics. Estimated premorbid intelligence, race, ethnicity, and the Marshall CT Class II (cisterns present) relative to normal CT were also independently associated with the global cognition score (see Table 3 and Fig. 3). Each tenfold change in the raw burden of severe IIC (Hz-hr) was associated with a 0.73 point (95% confidence interval 0.60–0.88 point; $p = 0.002$) lower global cognition score.

Table 3 Significant predictors of global cognitive score after adjusting for multiple variables

Variable	Estimate	Standard Error	*t* value	p value
Intercept	− 0.967	0.279	− 3.46	< 0.001
Study drug (vs. placebo)	0.037	0.103	0.36	0.722
Race (white)	0.324	0.115	2.81	0.006
Ethnicity (Hispanic)	0.423	0.120	3.52	0.001
Marshall CT II (cisterns present; shift < 5 mm)	− 0.195	0.093	− 2.09	0.040
NAART (verbal IQ)	0.338	0.092	3.67	< 0.001
Log (IIC burden)	− 0.316	0.097	− 3.24	0.002

CT, computed tomography, IIC, ictal-interictal continuum, NAART, North American Adult Reading Test

Similar models were constructed for each RBANS domain-specific subscore. In summary, the IIC burden was independently associated with immediate and delayed memory subscores and time to complete both TMT A and B (Supplementary Fig. 3).

Antiepileptic Drugs

In the cohort of patients with cognitive testing, 21 of 142 (15%) received no AEDs, whereas 53 (37%) received levetiracetam (LEV) and 68 (48%) received phenytoin or fosphenytoin (PHT). Of those who received AEDs, 9 of 121 (7%) received more than one AED; eight of these were on PHT plus one to three additional agents and one was on both LEV and valproic acid. In univariate analysis, patients who received LEV had higher IMPACT sum scores (median 2 [IQR 2–6] vs. median 2 [IQR 0–3] in those without AED or receiving PHT; $p = 0.02$), more IVH (18 of 53 [34%] vs. 2 of 21 [10%] in those

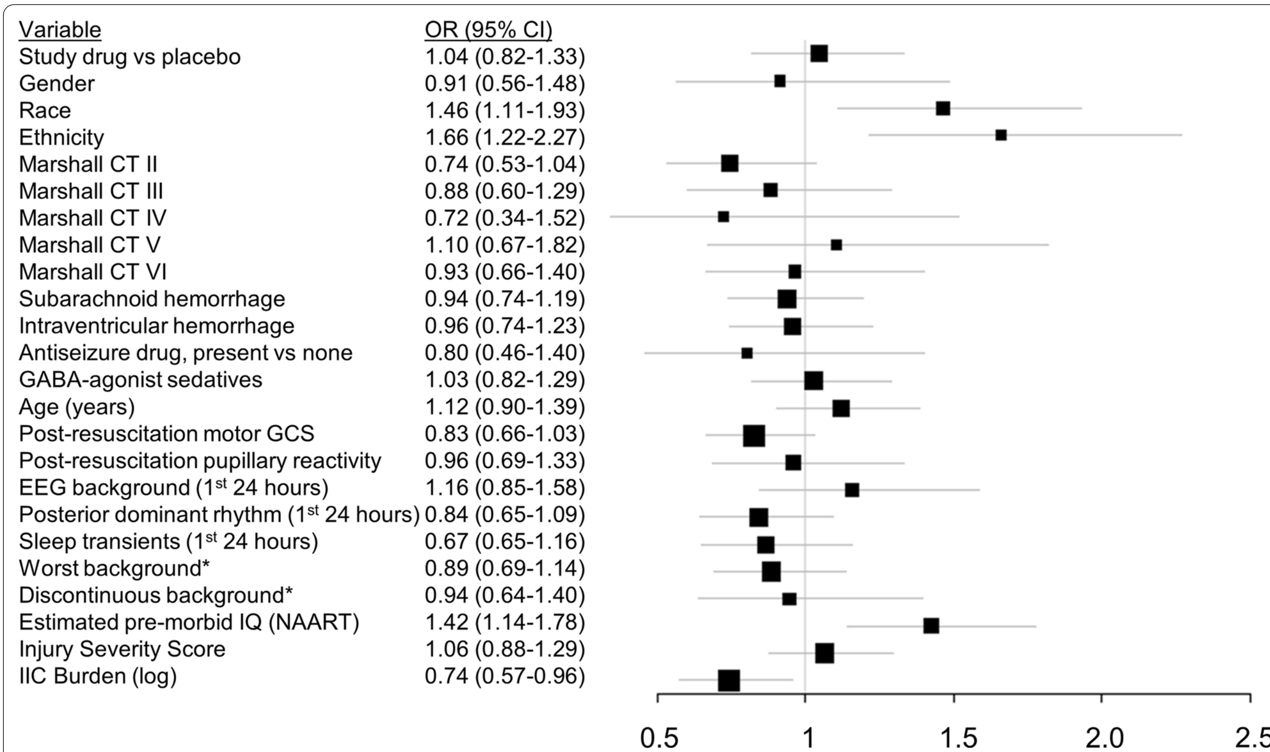

Fig. 3 Forest plot of multiple variable regression coefficients. Forest plot of the general linear regression coefficients for the predictors of the global cognition score. Coefficients were standardized by subtracting the mean and dividing by twice the corresponding standard deviations in order to directly compare coefficients between continuous and binary predictors. Controlling for age, sex, estimated premorbid intelligence, injury severity, radiographic variables, medications, and EEG background characteristics, the ictal-interictal (IIC) burden was independently associated with the global cognition score. *At anytime during monitoring. GABA, γ-aminobutyric acid, GCS, Glasgow Coma Scale, IIC, ictal-interictal continuum, NAART, North American Adult Reading Test

without AED and 11 of 68 [16%] in those receiving PHT; $p = 0.02$), and more subarachnoid hemorrhage (SAH) (40 of 53 [76%] vs. 8 of 21 [38%] in those without AED and 39 of 68 [57%] in those receiving PHT; $p = 0.01$). Of the 89 who underwent EEG monitoring, there were no differences between those on LEV vs. PHT in the proportion of patients with severe IIC patterns (6 of 33 [18%] vs. 7 of 43 [16%]; $p = 0.26$) or burden of severe IIC patterns (0.41 +/−1.71 Hz-hr vs. 0.46 +/−2.01 Hz-hr; $p = 0.70$). There were similarly no differences in the proportion with poor functional outcome (2 of 21 [15%] vs. 11 of 53 [33%] and 13 of 68 [30%], respectively; $p = 0.13$) or global cognition score (83.9 +/−16.5 vs. 81.8 +/−17.0 and 78.4 +/−13.4, respectively; $p = 0.27$) (Fig. 4).

Discussion

In patients with moderate to severe TBI, the burden of IIC patterns, including electrographic seizures and abnormal periodic or rhythmic discharges, was associated with worse cognitive outcome at 3 months in this post hoc analysis of a randomized controlled trial. Cognitive function, encompassing domains such as memory, attention, processing speed, and executive functioning, is commonly impaired after TBI and in our cohort of 142 survivors with TBI, the median global cognition score was nearly 1.5 SDs from the normative mean. We controlled for known confounders, including estimated premorbid intelligence and injury severity, and found that the burden of IIC inversely correlated with global cognition score and was independently associated with worse cognitive performance, particularly in memory, attention, and executive functioning. We found no relationship between the use of AEDs and either functional or cognitive outcome.

This study provides important evidence linking neuronal injury to clinically important patient outcomes. Prior research has clearly established the impact of

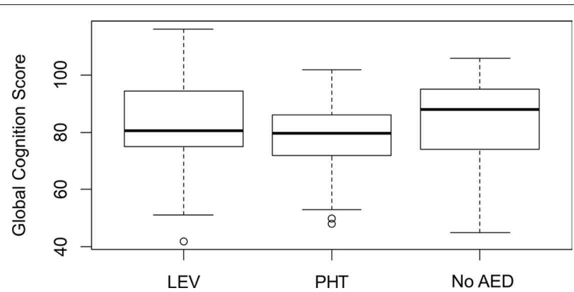

Fig. 4 Boxplot of antiepileptic drugs and global cognition score. Boxplot of the global cognition score (*y*-axis) across patients receiving each antiepileptic drug. There were no significant differences between each group and global cognition scores. AED, antiepileptic drug, LEV, levetiracetam, PHT, phenytoin and/or fosphenytoin

seizures on cortical physiology after primary brain injury. In a study of patients with SAH, electrographic seizures were found to exert similar physiologic effects as generalized tonic–clonic seizures [10]. Subsequent work has linked seizures with metabolic crisis [6], and increasingly frequent periodic discharges have been shown to increase oxygen extraction, decreasing the pool of dissolved tissue oxygen, or PbtO2 [23]. These findings help to explain the observation that seizures following TBI may be associated with hippocampal atrophy [7]; however, no study has definitively linked seizures following TBI with mortality or functional outcome.

One important reason for this lack of association between seizures and functional outcome is that the traumatic injury itself may contribute to functional disability far more than the subsequent development of seizures. In our prior study, we found that seizures and periodic discharges were indeed more common in those with more severe injury, explaining perhaps the lack of an independent association between seizures and functional outcome [3]. However, cognitive functions are thought to rely on distributed neuronal networks that may not be uniformly affected by moderate to severe injuries [24–26], whereas they may be disrupted by seizures leading to an impact on cognition [27]. For instance, the burden of seizure activity in patients with epilepsy has been linked with cognitive dysfunction, an *epileptic encephalopathy,* consisting of "…cognitive and behavioral impairments *above and beyond* what might be expected from the underlying pathology alone." [11, 28]. Here, we controlled for both injury severity and premorbid cognitive ability in order to understand the impact of acute seizures or abnormal periodic or rhythmic patterns above and beyond what might be expected after TBI.

The clinical importance of our results mirrors a recent study of 402 patients with diffuse SAH [29] in which a similar proportion (50 of 402 [12%]) had electrographic seizures lasting a median of 6 h. Each hour spent with seizures was associated with a decrease of 0.19 points on the Telephone Interview of Cognitive Status after controlling for level of education and ethnicity. We estimate that based on our model a decrease in global cognitive score reaching 0.5 SD, or 7.5 points, would be associated, for example, with prolonged periodic discharges at 2 Hz, totaling 36 h of an EEG monitoring period. Importantly, even a few hours of undetected or untreated seizures could be associated with cognitive recovery below the threshold of overt cognitive impairment, which is considered to be a score below 78 [30]. Whether and how acute seizures, or the subsequent development of epilepsy, modify cognitive *recovery* deserves further study.

Current guidelines recommend the use of PHT to prevent early seizures (level of evidence IIa[8]). In patients with TBI, randomized controlled trial data suggest that there are no differences in 1-year cognitive outcome between those who receive PHT vs. placebo [31]. However, after diffuse SAH, the use of PHT has been linked with worse cognitive outcome [32]. More recently, LEV has become the predominant AED used in neurocritical care [33]. Data from patients with focal intracerebral hemorrhage have suggested that patient-reported cognitive function after recovery might be adversely affected by LEV [34]. We found that rates of seizures or abnormal periodic or rhythmic patterns in patients with TBI were similar between those receiving LEV vs. PHT at a rate similar to prior studies [35, 36]. Further, we found no differences in functional or domain-specific cognitive outcomes related to the use of either AED compared with those in our cohort who did not receive AEDs.

This study is limited by its post hoc study design. There were no raw images to allow for an analysis of lesion burden and location, which precluded a detailed assessment of neuroanatomic correlates of cognition or EEG. However, we leveraged the prospective review of raw cEEG by a group of highly trained electroencephalographers and an independent validation and quantification by physicians trained in the critical care EEG terminology standards in order to minimize variability in the interpretation of EEG variables used as our primary exposure. Despite inclusion of EEG "wherever possible" in the study protocol, many patients did not undergo monitoring, potentially creating bias in the study cohort. Finally, the development of epilepsy or the occurrence of ictal or interictal patterns at the time of follow-up were not assessed, representing a potential confounder and major limitation to our conclusions.

We chose to include only those survivors able to perform cognitive testing in order to make specific inferences about the burden of IIC patterns and to leverage the population based normative RBANS index scores but recognize this strategy may limit generalizability in survivors unable to perform these assessments. Studies of posttraumatic cognition are challenging as a substantial proportion of the moderate to severe TBI population are unable to complete cognitive tests because of disability. In our study, these patients were older with more severe injuries. However, we found that global cognition scores in those able to complete cognitive testing were not significant different across a broad range of disability. Strategies to include all participants in future studies, including those unable to participate in standard neurocognitive testing, should include an ordinal ranking of cognitive and noncognitive outcome scores, including death and level of disability. Later time points might alternatively be used to maximize survivors who eventually regain the ability to perform cognitive tasks. Learning and memory deficits improve substantially in the first 4–6 months [24], and even in those with disorders of consciousness, more than two thirds regain consciousness within 6–8 weeks and exhibit significant increases in cognition that continue for up to 5 years [37]. Despite this continued long-term recovery, 60% of those with the most severe brain injuries continue to have cognitive deficits [26], with average global cognition scores in one series of patients 7 years following severe brain injury of 71 + / − 13 [19].

Conclusions

In conclusion, we found that the burden of seizures and abnormal periodic or rhythmic patterns after brain trauma were independently associated with objective measures of cognitive function after controlling for estimated premorbid intelligence and injury severity. Further study is warranted to determine whether the detection and treatment of IIC patterns in the initial postinjury period might lead to improvements in cognitive outcome for survivors of moderate to severe TBI.

Supplementary Information

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Author contributions

BF: conceived and designed the analysis, performed the analysis, wrote the paper, HL: contributed data and/or analysis tools, performed the analysis, critical revisions to the paper, MAM: contributed data and/or analysis tools, JAH: collected the data, critical revisions to the paper, LBN: critical revisions to the paper and relevant content expertise, MP: collected the data, contributed data or analysis tools, FCT: conceived and designed the initial study, collected the data, NZ: performed the analysis, JHK: contributed to performance of the analysis and relevant content expertise. Authorship requirements have been met, and the final manuscript was approved by all authors.

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Conflicts of interest
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